



UNIVERSITY OF SANTO TOMAS
COLLEGE OF SCIENCE
Academic Year 2026 - 2027



ADVENTURE WORKS PERFORMANCE ANALYTICS DASHBOARD
TECHNICAL DOCUMENTATION

By McReg Analytics

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ABOUT THE COMPANY

McReg Analytics is a data analytics company based in Ortigas, Pasig City, offering swift, efficient services to customers and striving to provide a bigger and brighter future. Our company guarantees professional, reliable services by transforming raw data into useful information that enables smarter decision-making. McReg Analytics assures customers that its team is highly competent, qualified, and dedicated to providing precise analytics. In addition, the company aims to assure customers that all information shared with McReg Analytics will be handled with the utmost discretion and confidentiality, processed securely, and used only for its intended purpose.

AdventureWorksDW2025 Documentation

Created Tables, Measurements, and Columns:

Created table “_Measures1”

Created measures:

Internet Sales = SUM(FactInternetSales[SalesAmount])

Profit Margin % = DIVIDE([Gross Profit], [Total Revenue], 0)

Reseller Sales = SUM(FactResellerSales[SalesAmount])

Revenue PY = CALCULATE([Total Revenue], SAMEPERIODLASTYEAR(DimDate[Date]))

Total Revenue = [Internet Sales] + [Reseller Sales]

Gross Profit = [Total Revenue] - [Total Cost]

Internet Sales % of Total =

DIVIDE(
[Internet Sales],
CALCULATE([Internet Sales], ALL(DimCustomer[NumberCarsOwned])),
0
)

Created table “_Measures2”

Created measures:

Sales Same Quarter Last Year \$ =

CALCULATE([Total Sales (All Channels)], DATEADD(DimDate[Date], -1, YEAR))



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Total Sales (All Channels) = [Total Internet Sales] + [Total Reseller Sales]
Total Internet Sales = SUM(FactInternetSales[SalesAmount])
Total Reseller Sales = SUM(FactResellerSales[SalesAmount])
Total Units Sold =
SUMX(FactInternetSales, FactInternetSales[OrderQuantity]) + SUMX(FactResellerSales,
FactResellerSales[OrderQuantity])
Profit Margin =
DIVIDE(
SUMX(FactInternetSales, FactInternetSales[SalesAmount] - FactInternetSales[TotalProductCost])
+ SUMX(FactResellerSales, FactResellerSales[SalesAmount] - FactResellerSales[TotalProductCost]),
[Total Sales (All Channels)]
)

Created table “_Measures3”

Created measures:

Top Continent by Revenue =

VAR TopCont = TOPN(1, VALUES(DimSalesTerritory[TerritoryDisplay]), [Total Sales (All Channels)],
DESC)

RETURN CONCATENATEX(TopCont, DimSalesTerritory[TerritoryDisplay], ", ")

Full Price Profit Margin % =

DIVIDE([Full Price Profit], CALCULATE([Total Internet Sales], FactInternetSales[PromotionKey] = 1))

Promotional Profit Margin % =

DIVIDE([Promotional Profit], CALCULATE([Total Internet Sales], FactInternetSales[PromotionKey] <>
1))

Promotional Avg Revenue Per Order =

CALCULATE(AVERAGE(FactInternetSales[SalesAmount]), FactInternetSales[PromotionKey] <> 1)

Full Price Avg Revenue Per Order =

CALCULATE(AVERAGE(FactInternetSales[SalesAmount]), FactInternetSales[PromotionKey] = 1)

Full Price Profit Margin % =

DIVIDE([Full Price Profit], CALCULATE([Total Internet Sales], FactInternetSales[PromotionKey] = 1))

Bikes-Only Customers =

VAR BikeBuyers =

CALCULATETABLE (VALUES (FactInternetSales[CustomerKey]), DimProduct[CategoryName] =
"Bikes")



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```
VAR AccessoryBuyers =  
    CALCULATETABLE ( VALUES ( FactInternetSales[CustomerKey] ), DimProduct[CategoryName] =  
"Accessories" )  
VAR BikesOnlyBuyers = EXCEPT ( BikeBuyers, AccessoryBuyers )  
RETURN  
    COUNTROWS ( BikesOnlyBuyers )  
    Bikes-Only Share % =  
DIVIDE ( [Bikes-Only Customers], [Total Bike Buyers] )
```

Created table "Channel"

Created measures:

Reseller Gross Margin % =

```
DIVIDE ( [Reseller Sales Amount] - [Reseller Cost], [Reseller Sales Amount] )
```

Internet Gross Margin % =

```
DIVIDE ( [Internet Sales Amount] - [Internet Cost], [Internet Sales Amount] )
```

Total Profit Summary =

```
VAR TotalProfit = [Combined Profit]
```

```
VAR ResShare = [Reseller Profit Share %]
```

```
VAR IntShare = [Internet Profit Share %]
```

RETURN

```
FORMAT ( TotalProfit, "$#,##0,, ""M"" )
```

```
& " (Reseller " & FORMAT ( ResShare, "0%" )
```

```
& " • Internet " & FORMAT ( IntShare, "0%" ) & ")"
```

Total Revenue Summary =

```
VAR TotalRev = [Combined Sales Amount]
```

```
VAR ResShare = [Reseller Revenue Share %]
```

```
VAR IntShare = [Internet Revenue Share %]
```

RETURN

```
FORMAT ( TotalRev, "$#,##0,, ""M"" )
```

```
& " (Reseller " & FORMAT ( ResShare, "0%" )
```

```
& " • Internet " & FORMAT ( IntShare, "0%" ) & ")"
```

Channel Revenue =

SWITCH (

```
    SELECTEDVALUE ( Channel[Channel] ),
```

```
    "Reseller", [Reseller Sales Amount],
```

```
    "Internet", [Internet Sales Amount],
```



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```
BLANK ()
)
    Top Country (Internet) Combined =
VAR CountryTable =
    ADDCOLUMNS (
        ALLSELECTED ( DimGeography[Country] ),
        "@Rev", [Internet Sales Amount]
    )
VAR TopRow =
    TOPN ( 1, CountryTable, [@Rev], DESC )
VAR TopCountry = MAXX ( TopRow, DimGeography[Country] )
VAR TopRev = MAXX ( TopRow, [@Rev] )
RETURN
    TopCountry & " " & FORMAT ( TopRev, "$#,##0,.""M"" )
    Top Reseller Combined =
VAR ResellerTable =
    ADDCOLUMNS (
        ALLSELECTED ( DimReseller[ResellerName] ),
        "@Rev", [Reseller Sales Amount]
    )
VAR TopRow =
    TOPN ( 1, ResellerTable, [@Rev], DESC )
VAR TopName = MAXX ( TopRow, DimReseller[ResellerName] )
VAR TopRev = MAXX ( TopRow, [@Rev] )
RETURN
    TopName & " " & FORMAT ( TopRev, "$#,##0,.""M"" )
    Channel Revenue =
SWITCH (
    SELECTEDVALUE ( Channel[Channel] ),
    "Reseller", [Reseller Sales Amount],
    "Internet", [Internet Sales Amount],
    BLANK ()
)
Channel Gross Margin % =
VAR Rev = [Channel Revenue]
VAR Cost = [Channel Cost]
RETURN DIVIDE ( Rev - Cost, Rev )
    Internet Units Sold = SUM ( FactInternetSales[OrderQuantity] )
    Reseller Units Sold = SUM ( FactResellerSales[OrderQuantity] )
    Sales Reason Order Count =
DISTINCTCOUNT ( FactInternetSalesReason[SalesOrderNumber] )
```



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```
Internet Avg Units Per Order = DIVIDE ( [Internet Units Sold], [Internet Order Count] )
Top Performer =
VAR EmployeeTable =
    ADDCOLUMNS (
        ALLSELECTED ( DimEmployee[FullName] ),
        "@Rev", [Reseller Sales Amount]
    )
VAR TopRow =
    TOPN ( 1, EmployeeTable, [@Rev], DESC )
VAR TopName = MAXX ( TopRow, DimEmployee[FullName] )
VAR TopRev = MAXX ( TopRow, [@Rev] )
RETURN
    TopName & " " & FORMAT ( TopRev, "$#,##0,, ""M"" )
```

```
Created columns:
Channel = DATATABLE(
    "Channel", STRING,
    {
        {"Reseller"},
        {"Internet"}
    }
)
```

From table “DimDate”

```
Created column:
YearQuarter = "Q" & DimDate[CalendarQuarter] & " " & DimDate[CalendarYear]
Season =
SWITCH(
    TRUE(),
    DimDate[MonthNumberOfYear] IN {12, 1, 2}, "Winter",
    DimDate[MonthNumberOfYear] IN {3, 4, 5}, "Spring",
    DimDate[MonthNumberOfYear] IN {6, 7, 8}, "Summer",
    "Fall"
)
QuarterYear = "Q" & DimDate[CalendarQuarter] & " " & DimDate[CalendarYear]
```



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From table "FactResellerSales"

Created measures:

Reseller Discount Rate =
DIVIDE ([Reseller Discount], [Reseller Gross Amount])
Total Discount Given =
SUMX(FactInternetSales, FactInternetSales[SalesAmount] * RELATED(DimPromotion[DiscountPct]))
Reseller Discount Rate =
DIVIDE ([Reseller Discount], [Reseller Gross Amount])
Reseller Sales Amount = SUM (FactResellerSales[SalesAmount])

From table "FactInternetSales"

Created measures:

Internet Sales Amount = SUM (FactInternetSales[SalesAmount])

From table "DimPromotion"

Created columns:

PromotionStatus = IF (DimPromotion[PromotionName] = "No Discount", "Full Price",
"Promotional")

From table "FactSalesQuota"

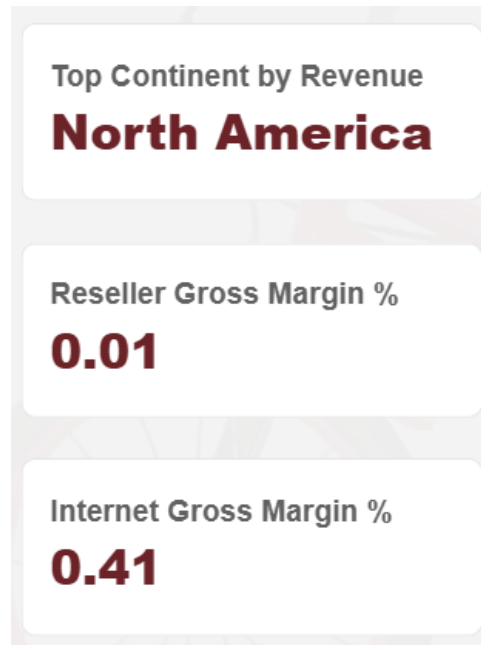
Created measures:

Total Sales Quota = SUM (FactSalesQuota[SalesAmountQuota])
Quota Attainment_1 % =
DIVIDE ([Reseller Sales Amount], [Total Sales Quota])

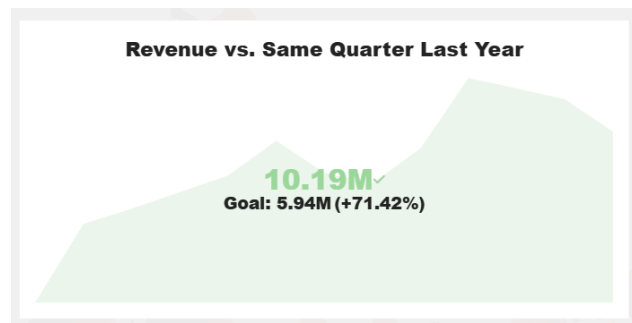
EXECUTIVE OVERVIEW



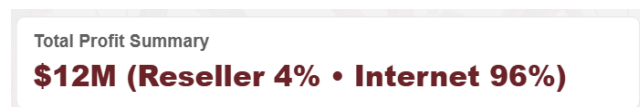
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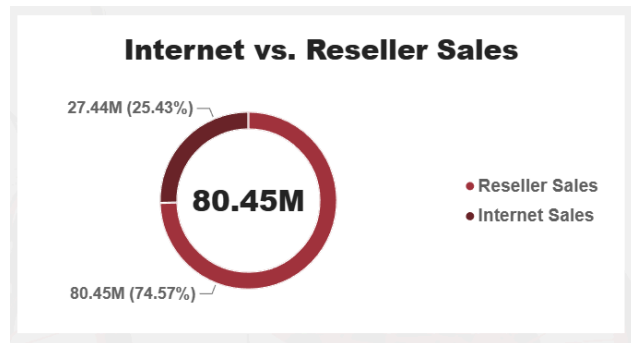
- Used a slicer



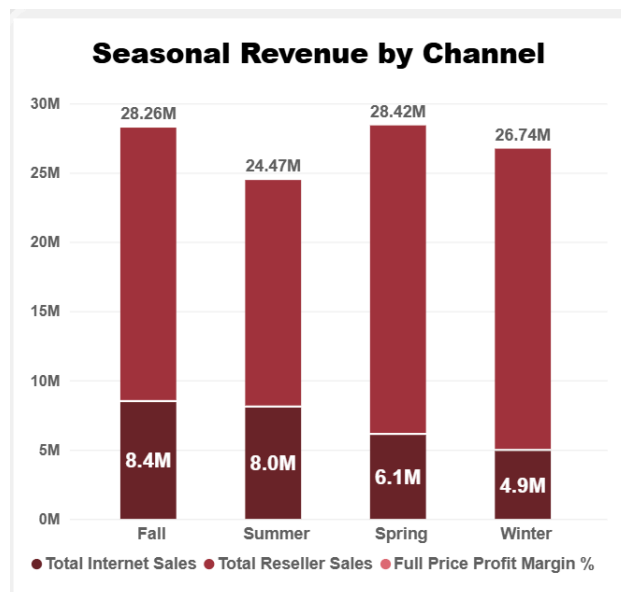
- Used KPI



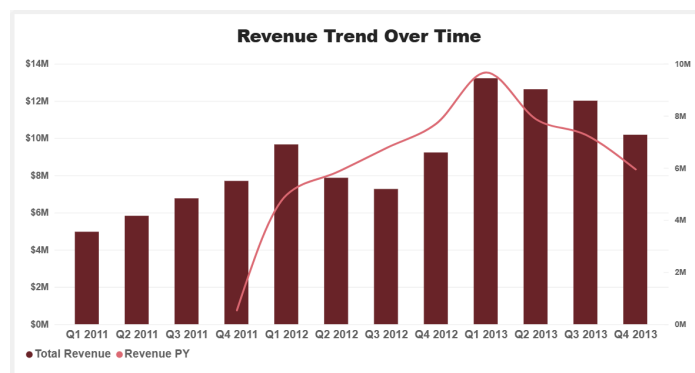
- Used card



- Used a donut chart



- Used a stacked column chart



- Used a combined line and column chart



- Used a filled map

CHANNEL PERFORMANCE



- Used a slicer

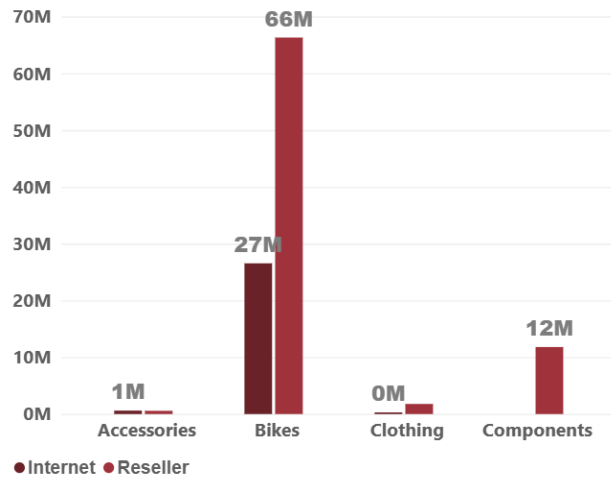


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Channel Revenue

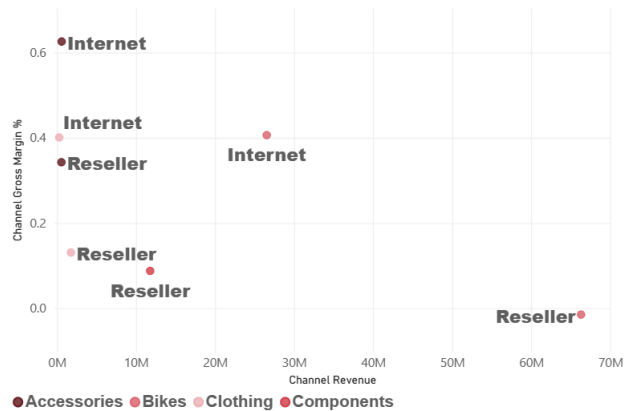
by CategoryName and Channel



- Used a clustered column chart

Channel Revenue and Channel Gross Margin %

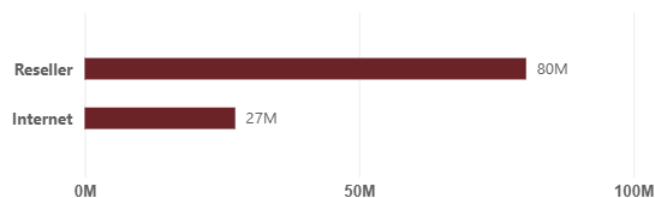
by Channel and CategoryName



- Used a scattered plot

Channel Revenue

by Channel





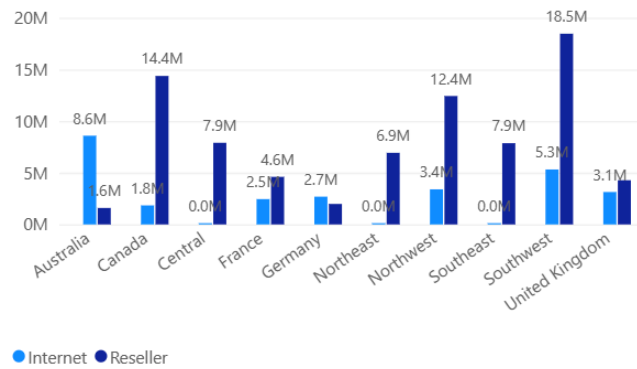
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- Used a bar chart

Channel Revenue

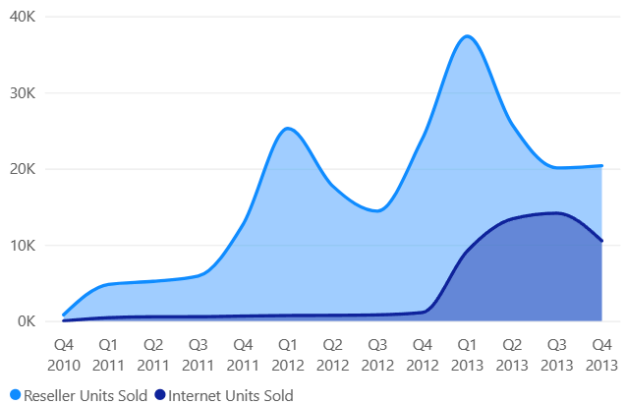
by SalesTerritoryRegion and Channel



- Used a clustered column chart

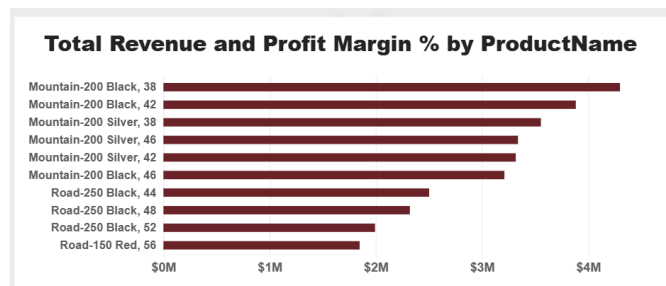
Reseller Units Sold and Internet Units Sold

by QuarterYear



- Used a stacked area chart

PRODUCT PERFORMANCE





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- Used a bar chart

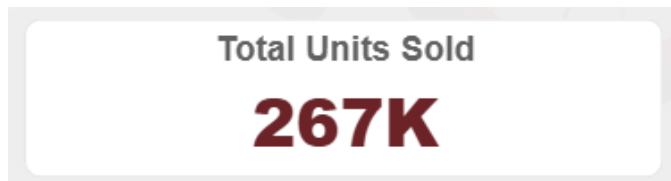


- Used a line and stacked column chart

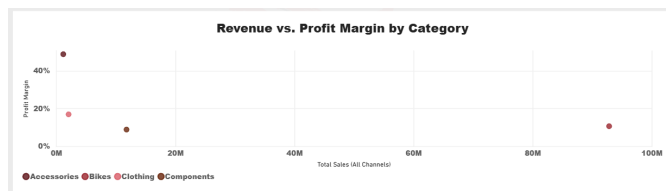
Category & Subcategory Profitability Breakdown

CategoryName	Total Revenue	Gross Profit	Profit Margin %
Accessories	\$1,175,903.19	\$574,274.48	48.84%
Bikes	\$92,846,754.40	\$9,793,773.83	10.55%
Clothing	\$2,067,484.95	\$348,559.40	16.86%
Components	\$11,799,076.66	\$1,032,966.48	8.75%
Total	\$107,889,219.19	\$11,749,574.19	10.89%

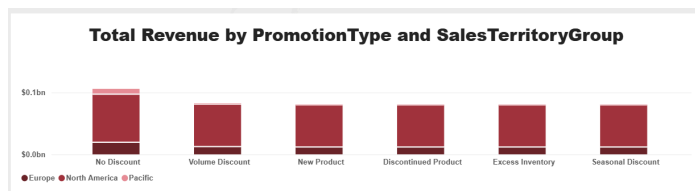
- Used a table



- Used a slicer

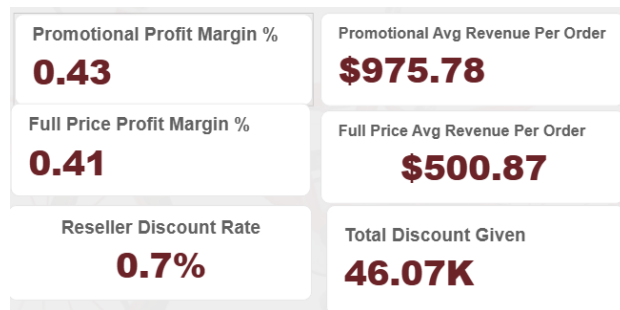


- Used a scatter chart

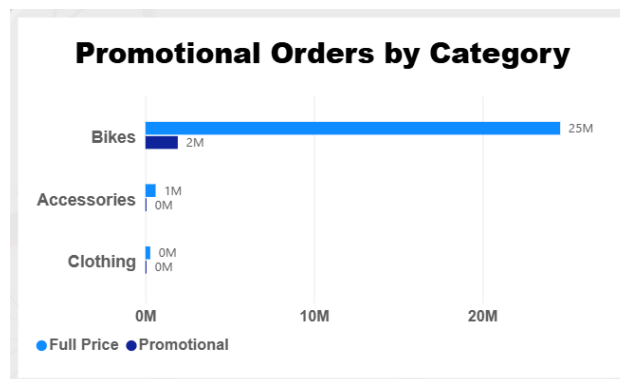


- Used a line and stacked column chart

MARKETING PERFORMANCE

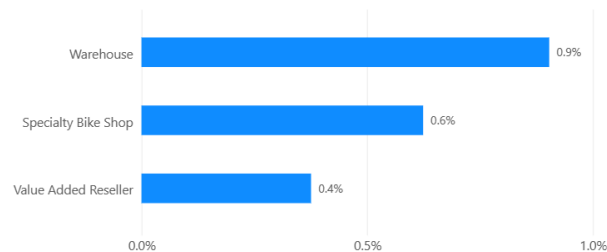


- Used a slicer



- Used a clustered bar chart

Reseller Discount Rate
by BusinessType



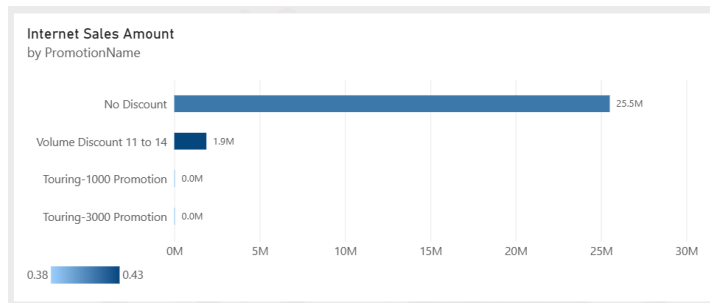
- Used a bar chart



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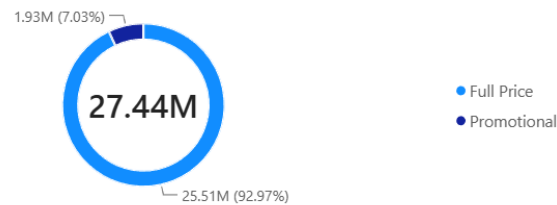
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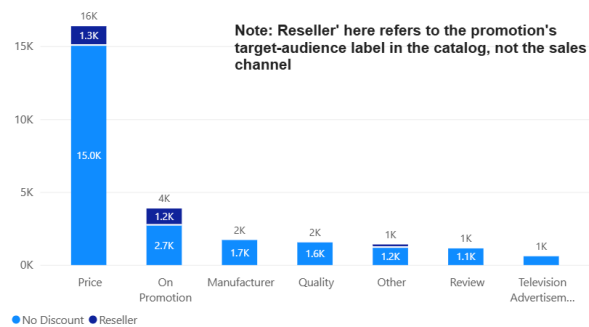
- Used a bar chart

Internet Sales Amount
by PromotionStatus



- Used a donut chart

Sales Reason Order Count
by SalesReasonName and PromotionCategory



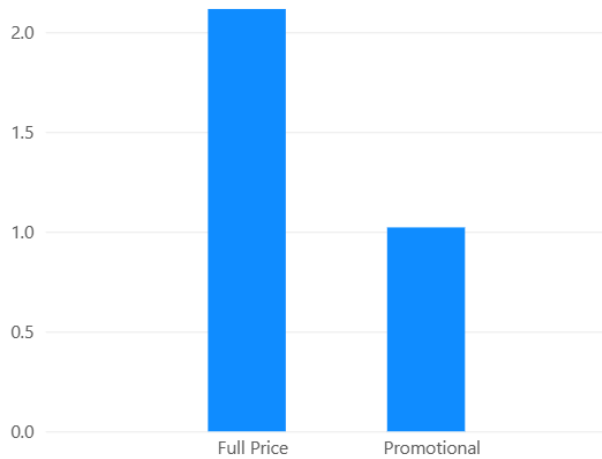
- Used a stacked column chart



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Internet Avg Units Per Order
by PromotionStatus



- Used a column chart

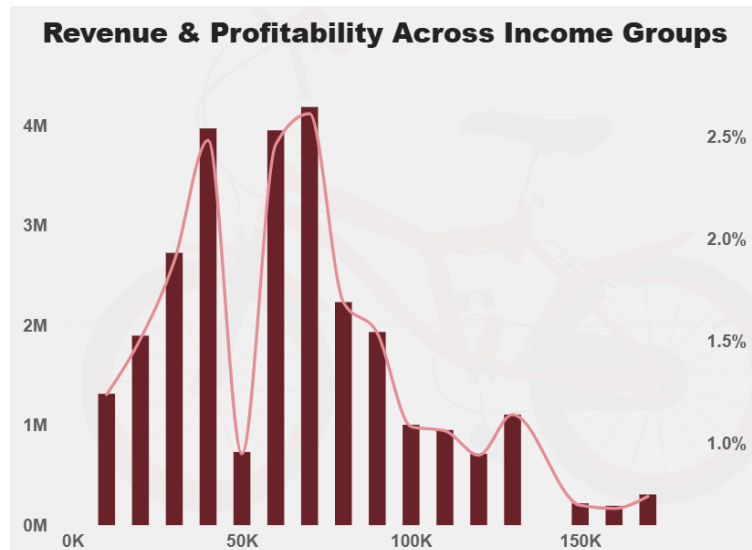
CUSTOMER INSIGHTS



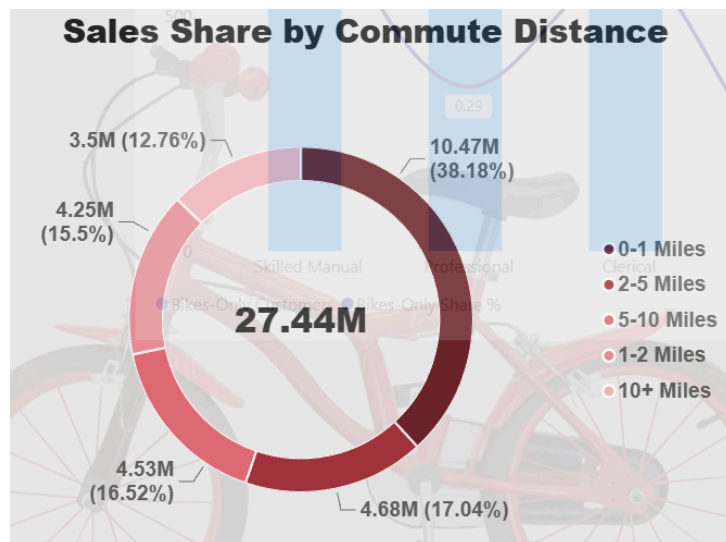
- Used a bar chart



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- Used a combined line and stack column chart



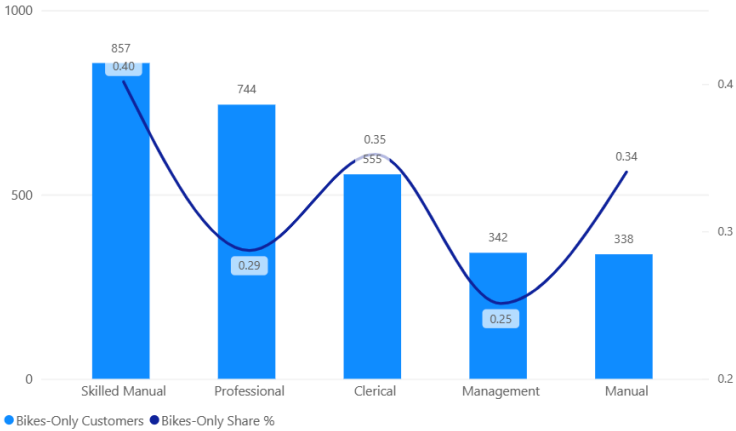
- Used a donut chart



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Bikes-Only Customers and Bikes-Only Share %
by Occupation



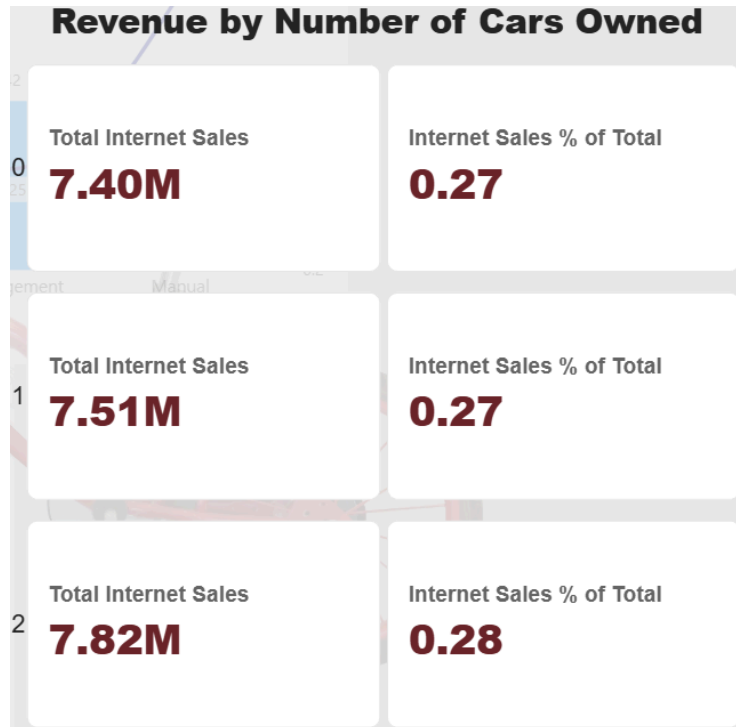
- Used combined line and stacked column chart

FullName	Occupation	Sum of YearlyIncome	CommuteDistance	Sum of NumberCarsOwned	Sum of NumberChildrenAtHome	Inte
Abigail Morris	Clerical	20000	1-2 Miles	2	0	
Alex Collins	Clerical	20000	1-2 Miles	2	0	
Cody Sanders	Clerical	20000	10+ Miles	4	0	
Ebony Gill	Clerical	40000	1-2 Miles	1	0	
Lauren Miller	Clerical	40000	1-2 Miles	1	0	
Allison Evans	Management	60000	10+ Miles	0	1	
Bradley Rai	Management	100000	10+ Miles	3	5	
Erin Sanders	Management	70000	5-10 Miles	2	0	
Abigail Bennett	Professional	30000	5-10 Miles	2	0	
Ashley Jones	Professional	60000	10+ Miles	2	1	
Dalton Clark	Professional	60000	1-2 Miles	2	0	
Dylan Taylor	Professional	30000	5-10 Miles	2	0	
Muster Miller	Professional	20000	1-2 Miles	2	0	
Total		1200000	1-2 Miles	48	13	

- Used a table



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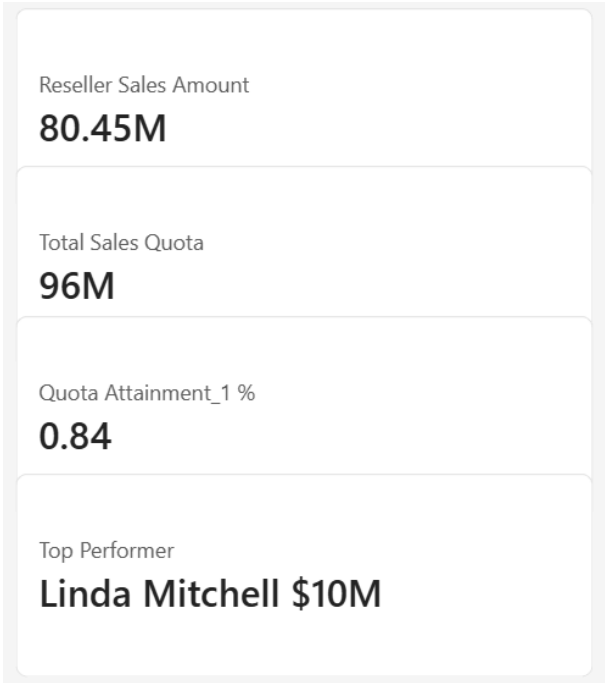


- Used a slicer

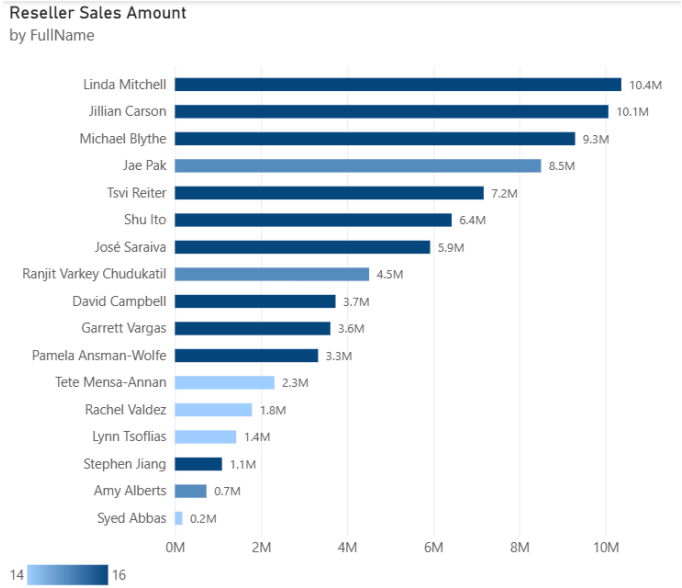
EMPLOYEE EVALUATION



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- Used slicer



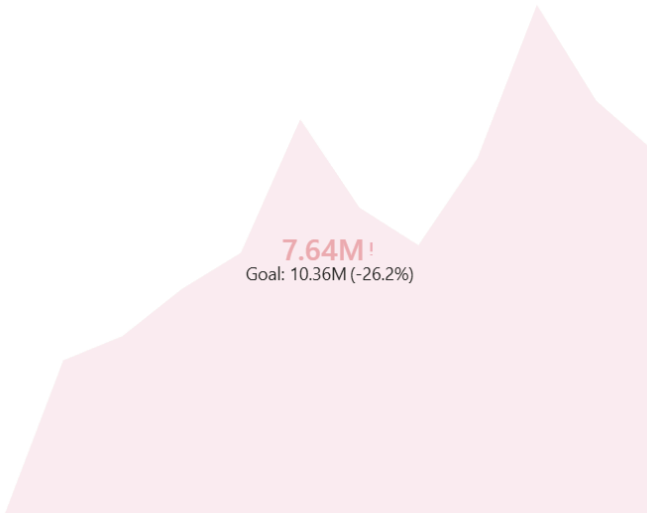
- Used a bar chart



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Reseller Sales Amount and Total Sales Quota
by QuarterYear

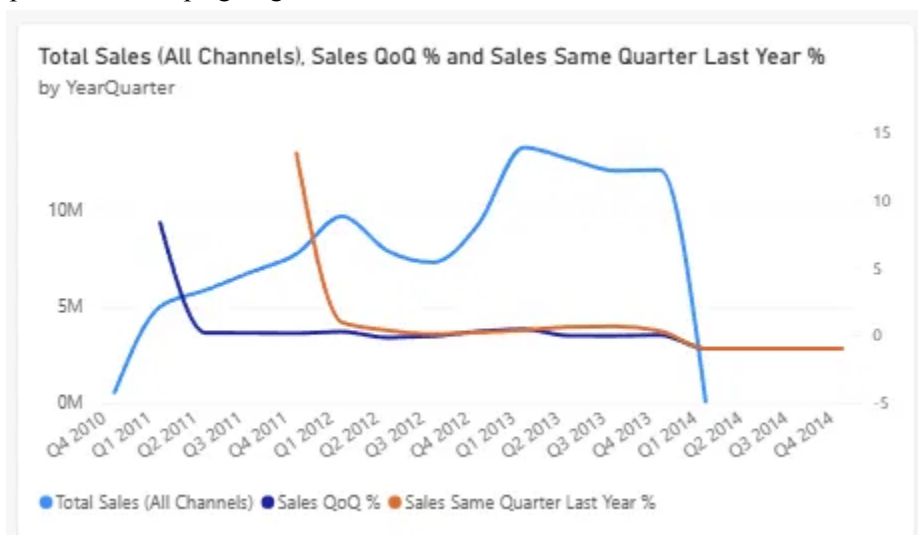


- Used a KPI

Changes Made during exploration:

#1

Upon trying to do a line chart, theres an abnormal drop of total sales, whilst the sales quarter over quarter still keeps going.



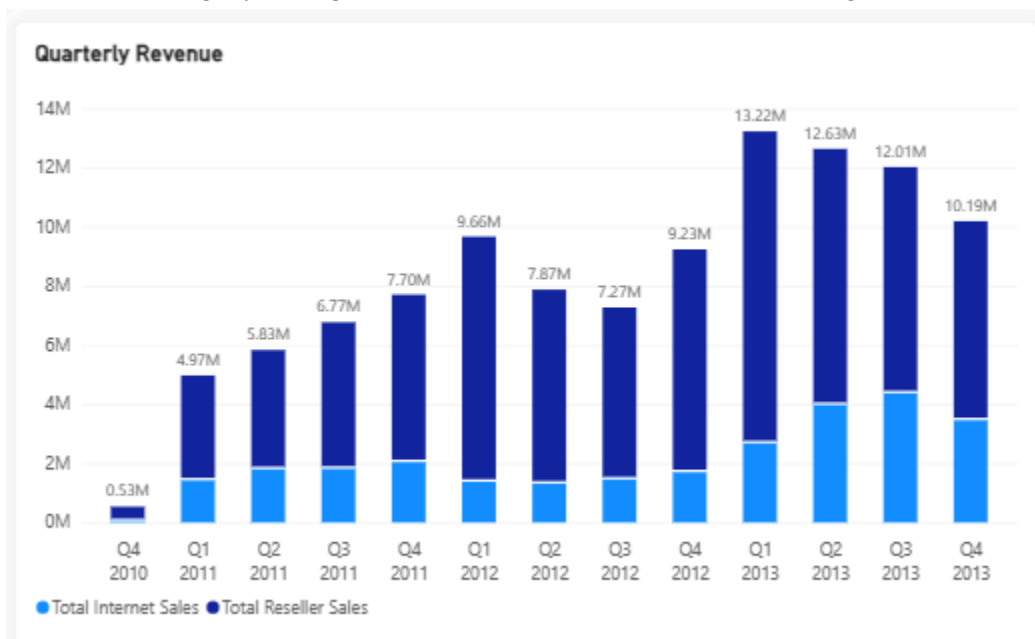
We did a double check on sql

```
1 SELECT MAX(OrderDate) FROM FactInternetSales;
2 SELECT MAX(OrderDate) FROM FactResellerSales;
```

Results	
(No column name)	
1	2014-01-28 00:00:00.000
(No column name)	
1	2013-11-29 00:00:00.000

Query executed successfully. localhost (17.0 RTM) MEH\CO (63) AdventureWorksDW2025 00:00:00 Row: 1, Col: 1 2 rows

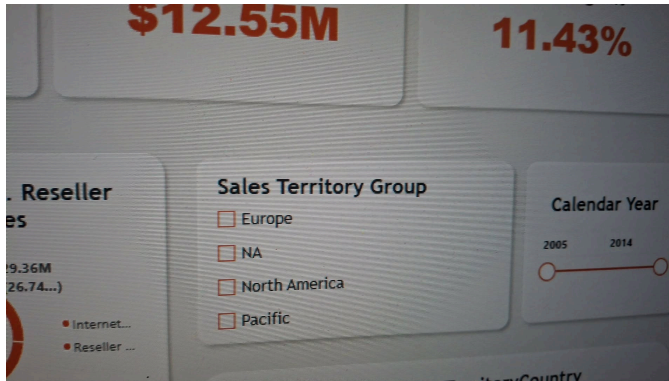
We decided to do a simple filter instead, on the table “DimDate” specifically at “Date”. We did an advanced filtering, by setting the condition “is on or before”, then setting the date 11/30/2013.



Reason: Since Internet Sales reported data was until January 2014 while Reseller Sales reported data was only until November 2013. Hence, to ensure fairness of comparing between both channels, only the date where they're both complete should be shown and analyzed which is November 2013.

#2

Upon making the slicer, we have encountered a NA at the territory group.



We did a double check on sql

```
6 SELECT st.SalesTerritoryGroup, COUNT(*) AS ResellerCount
7 FROM DimSalesTerritory st
8 LEFT JOIN DimReseller r ON r.GeographyKey = st.SalesTerritoryKey
9 GROUP BY st.SalesTerritoryGroup;
```

	SalesTerritoryGroup	ResellerCount
1	Europe	5
2	NA	3
3	North America	11
4	Pacific	1

Since theres 3 resellers to NA, small but valuable, so therefore cannot be dismissed. Therefore, under the table titled DimSalesTerritory, a column was created titled SalesTerritoryGroup (Display) to rename the NA to Other/Unassigned instead.

#3

Upon making a stacked column chart for the classes, we encountered a NULL

Total Sales (All Channels) and Product Profit Margin %
by Class





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We did a double check in sql

```
12 SELECT Class, COUNT(*) AS ProductCount, SUM(sales.SalesAmount) AS Revenue
13 FROM DimProduct p
14 LEFT JOIN FactInternetSales sales ON sales.ProductKey = p.ProductKey
15 GROUP BY Class;
16
17 SELECT DISTINCT Class FROM DimProduct;
```

	Class	ProductCount	Revenue
1	H	10984	22340591.8294
2	NULL	39186	861765.70
3	L	5299	2133761.2238
4	M	5377	4022558.4675

NULL class is the 2nd most ordered product, hence it cannot be removed. On the table titled "TerritoryDisplay", we added a new column titled "ClassDisplay" to reame NULL as "Accessories/Components (Unclassified)" instead.
